

AGENTPROF: Semantic Profiling for AI Agents

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Abstract

AI agents orchestrate long-running activities across users, tools, and system resources, producing growing populations of execution histories. Developers need population-level answers about where failures concentrate, which workflows trigger unsafe effects, and which task categories consume the most budget. Unlike stable code paths, natural-language intent lacks shared identity, and native execution nesting does not provide a reusable cross-run hierarchy of semantic responsibility. Existing agent tools trace runs, group metadata, and aggregate metrics, but do not combine source-linked system effects, conserved additive measures, and selectable query-time operation stacks over the same histories. Agent observability needs profiling, not only debugging. AGENTPROF realizes this insight through a *semantic operation stack model* with uniform weighted *operations*, query-time *operation stacks*, pluggable field and stack construction, label-free recurrence over visible action transitions, and pprof-compatible output. We evaluate AGENTPROF on a 20-task real Codex capture suite, 405 human-staged CodeTraceBench trajectories, three complete public localization benchmarks, 287 OSWorld-Human sessions, and four public cost workloads. Within the declared process/tool scope, AgentSight capture and joining reach 100.0% precision and 96.6% recall, reject 1,629/1,629 concurrent-control effects. Label-free recurrence raises ordinary B^3 F1 against human stages from 0.541 to 0.649, and semantic grouping improves MAP over raw action in all three localization benchmarks. Evaluated task-family and action backends reach 0.695 and 0.498 macro-F1, label-free recurrence reaches 0.680 boundary F1 and 0.786 B^3 F1 on OSWorld-Human, and AGENTPROF constructs a 27,765-operation profile in 1.17 s.

Introduction

AI agents (Yang et al. 2024; OpenAI 2025; Anthropic 2025; Xie et al. 2024) orchestrate multi-step activities across a high-level intent layer (prompts, LLM calls, and tool invocations) and a low-level system-effects layer (process spawns and file and network I/O). As agents evolve from short scripts into complex systems, production teams accumulate many trajectories.

Developers lack population-level answers about agent quality, safety, and cost. Which categories of tasks consume

the most token budget? Where do failures concentrate across workflows? Which behavioral patterns trigger unsafe system effects? At scale, manual trajectory inspection is slow and expensive (Lù et al. 2025), while LLM judging requires a separate evaluator pass per trajectory (Lù et al. 2025; Zheng et al. 2023).

Agent trajectories lack two structures needed for cross-run profiling. Responsibility lies in semantic categories such as task intent and tool operation rather than stable function names. Natural-language prompts provide no common identifier for shared intent. Native execution nesting may exist, but it does not provide a reusable cross-run hierarchy of semantic responsibility. A prompt sequence may represent multiple tasks, while one task may belong to a larger task. Thus adapting profiling requires stable semantic identifiers and an attribution hierarchy that does not depend on a run-time stack.

Existing agent observability lacks this profiling abstraction. Tools provide rich tracing, metadata aggregation, and population dashboards (LangChain 2024; Langfuse 2024; OpenTelemetry 2024; Datadog 2025), and some derive hierarchical categories across traces and aggregate latency, cost, or evaluation metrics (LangChain 2026; Datadog 2026), while recent research canonicalizes actions into process profiles or cross-run graphs for monitoring and recovery (Shu et al. 2026; Liu et al. 2026a; Zhong, Saxena, and Raghunathan 2026; Nian et al. 2026). These systems establish semantic grouping, metric rollups, and cross-run analysis. They do not combine source-linked agent and system effects, conservation of arbitrary additive measures, and selectable query-time operation stacks over the same histories.

Agent observability needs profiling, not only debugging. Profiling transfers by projecting resources onto stable semantic tags in an attribution hierarchy. Our *semantic operation stack model* provides this structure through operations and operation stacks. *Operations* are uniform records with string fields and additive measures that represent prompts, LLM calls, tool invocations, and system effects without type-specific objects. *Operation stacks* replace the runtime call stack by ordering selected fields. Changing those fields attributes the same operations by task category, execution phase, session, or action type (Figure 2).

We implement this model in AGENTPROF, an offline profiler that compiles local agent trajectories into pprof-

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compatible profiles (Google 2024) (Figure 1). *Intent attribution* derives stable, low-cardinality tags from natural-language fields via regex rules, a local LLM tagger, or unsupervised clustering, replacing function-name identity. *Stack construction* uses either selected fields or label-free recurrence over visible cross-session action transitions to replace runtime nesting.

On 20 real Codex tasks, scoped AgentSight (Zheng et al. 2025) capture and joining reach 100.0% precision and 96.6% recall. Across all 405 reconstructable failed CodeTraceBench trajectories (Li et al. 2026), label-free recurrence raises ordinary B³ F1 against human stages from 0.541 to 0.649 over raw action. Across three complete public workloads, MAP over target-bearing trajectories rises from 0.773 to 0.789, 0.281 to 0.452, and 0.121 to 0.230, respectively (Fan et al. 2026; Wang et al. 2026; Chen et al. 2026a). On 287 OSWorld-Human development sessions (Abhyankar, Qi, and Zhang 2025), label-free recurrence reaches 0.680 boundary F1 and 0.786 B³ F1. On the AgentBoard and ASE populations, task-family and action backends reach 0.695 and 0.498 macro-F1, respectively (Ma et al. 2024; Bouzenia and Pradel 2025). On four public workloads, AGENTPROF constructs a 27,765-operation profile in 1.17 s.

This paper makes three contributions:

1. **Semantic operation stack model.** Uniform operations and query-time stacks supply agent observability’s missing profiling layer (Background and Motivation; Design).
2. **System: AGENTPROF.** Pluggable intent attribution and stack construction produce pprof-compatible profiles (Implementation).
3. **Evaluation.** Semantic profiles preserve linked effects and improve attribution and problem ranking over raw action. Field evaluations cover task partitions, the integrated task-family path, a standalone action-label adapter, and session-held-out groups. AGENTPROF profiles 27,765 operations in 1.17 seconds (Evaluation).

Background and Motivation

LLMs and AI Agents

A typical agent trajectory interleaves two layers of activity (Yang et al. 2024; OpenAI 2025; Anthropic 2025; Zheng et al. 2025). At the intent layer, an agent receives prompts, issues LLM calls, and invokes tools for code execution, search, and file editing. Tool invocations trigger lower-layer process, file, and network effects (Zheng et al. 2025). Trajectories repeat this cycle, and teams accumulate many trajectories.

System Profiling

Profiling aggregates measurements to attribute resources to responsible entities rather than debugging a single execution. Traditional profilers sample execution, attach each sample to its call stack, and fold identical stacks (Google 2024; Gregg 2011). Flame graphs (Gregg 2011) and pprof (Google 2024) call graphs visualize aggregate cost by width or node size, while Pivot Tracing (Mace, Roelke, and Fonseca 2015) dynamically selects and groups measurements across causally

related events. These tools start from stable code identifiers and runtime stacks, although pprof can also promote tagroot/tagleaf values to pseudo-frames. Agent profiles must instead derive stable semantic fields and connect them to downstream effects.

Challenges for Agent Profiling

Existing tools (LangChain 2024; Langfuse 2024; OpenTelemetry 2024) record agent events as per-execution span trees for single-run debugging. AgentSight (Zheng et al. 2025) bridges intent and system-effect layers by correlating extended Berkeley Packet Filter (eBPF) system events with intercepted LLM traffic, but the resulting recordings still lack aggregate profiling.

Agent profiling faces two challenges. First, responsibility lies in semantic categories such as task intent and workflow phase, not code paths. Standards represent application-supplied span attributes (OpenTelemetry 2024; Arize AI 2024), but natural-language histories lack stable categories because different prompts or tools can express the same intent.

Second, agent trajectories lack a runtime call hierarchy for semantic attribution. In CPU profiling, the stack attributes malloc’s cost simultaneously to parse, process, and main. A prompt sequence may contain one or several tasks within a larger workflow, yet no runtime mechanism marks its boundaries or attribution granularity. RQ1 evaluates how this missing hierarchy affects attribution, while RQ3 evaluates constructed groups and boundaries.

Design Requirements for Agent Profiling

Traditional profiling relies on three mechanisms supplied automatically by the call stack. Agent profiling requires all three without a call stack:

- D1: Cross-layer resource projection.** The profiler must connect resource consumption from file I/O and process spawns to the responsible prompts and LLM calls.
- D2: Stable tags for folding.** Without stable function names, the profiler must derive short, repeatable tags from natural language before folding.
- D3: Hierarchical attribution.** Because span trees do not encode semantic responsibility through function-call nesting, the profiler must construct attribution hierarchies from the data.

Design

The design maps D1–D3 to ingestion, tags, and operation stacks. For D1, source-specific ingestion links each system effect to its intent, and operations preserve inherited semantic fields for projection and folding. For D2, intent attribution derives stable tags from natural-language fields. For D3, *operation stacks* replace runtime stacks with query-time field projections, supporting multiple attribution granularities over the same data.

The profiling pipeline has four stages: (1) parse raw trajectories into operations, (2) enrich operations via intent attribution or mapping rules, (3) project each operation onto

a stack frame sequence chosen at query time, and (4) fold operations with identical stacks, summing their weights.

Semantic Operation Stack Model

AGENTPROF represents each activity from the intent and system-effect layers as an *operation*, a weighted record with string fields and additive measures. Each operation carries string fields (`project`, `agent`, `session`, `prompt_tag`, `kind`, `model`, `path`, `domain`, `status`, ...) and additive measures (token count, duration, event count). The same schema represents prompts, LLM calls, tool invocations, file reads, GUI clicks, and process events. Populated fields distinguish these operation types. When a tool invocation triggers system effects, ingestion propagates its semantic fields to the resulting operations, allowing downstream resource weights to fold under the responsible intent.

An *operation stack* replaces the runtime call stack to provide hierarchical attribution for agent trajectories. In CPU profiling, a call stack determines which code paths share responsibility for a resource cost. In AGENTPROF, an ordered list of fields $[f_1, f_2, \dots, f_k]$ projects each operation o to a frame sequence $\langle o.f_1, o.f_2, \dots, o.f_k \rangle$, and operations with identical sequences are merged, summing their weights. Unlike a call stack, the fields are chosen at query time, not determined by execution. The same operations can therefore support a flat task summary, a per-session tree, a semantic phase/action hierarchy, or a trajectory organized by the dataset’s own annotations. Field selection changes attribution but not the data. Sessions and spans are optional fields, not hierarchy levels. The same data supports debugging views with *session* and aggregate profiling views without that field.

We formalize each selectable profile as a view (φ, σ, w) . For operation o , the predicate $\varphi(o)$ decides inclusion, an ordered field list $[f_1, \dots, f_k]$ defines $\sigma(o) = \langle o.f_1, \dots, o.f_k \rangle$, and $w(o) \geq 0$ supplies its weight. Users choose the stack function independently of the predicate and weight function. Four built-in resource choices define φ and w for common questions: (1) *tokens* weights LLM calls by token count, (2) *time* weights timed operations by duration, (3) *files* weights file operations by event count, and (4) *network* weights network operations by event count.

Attribution and Stack Construction

Intent attribution. Intent attribution maps natural-language fields to short, stable, repeatable tags, supplying the identifiers that traditional profiling obtains from function names. The framework supports three pluggable backends. Explicit rules provide repeatable user control, a local model tags natural-language fields, and unsupervised grouping discovers candidate categories that developers can encode as rules. Structured trajectories use the same interface to derive fields from existing action and quality attributes. All backends emit ordinary operation fields, keeping the stack model independent of how a tag was obtained.

Stack construction. With an ordered field list, stack construction directly projects the available fields. When users enable automatic construction, AGENTPROF induces operation

stacks without a user-specified field list through *label-free recurrence*. Visible actions are action-field values available before target annotations are read. The constructor uses adjacent visible-action recurrence across reference sessions, so weak or unseen transitions start segments and recurring transitions continue them. Each segment’s run-length-compressed action sequence becomes a reusable stack-frame value. The derived group and stack-frame values are ordinary operation fields, so the stack function can compose them with task, phase, action, or session fields at query time. The constructor can learn from a separate label-free operation corpus, so it does not use the target session’s problem annotations.

Implementation

AGENTPROF is an approximately 9.8K-line offline Rust command-line interface that implements the semantic operation stack model (Figure 1).

Input reconstruction. AGENTPROF reads Codex and Claude Code local JSON Lines (JSONL) histories and operation JSONL. For native histories, it reconstructs prompts, LLM calls, and tool invocations. Because the current CLI does not read AgentSight (Zheng et al. 2025) recordings directly, those recordings pass through a source-specific adapter. For AgentSight recordings, the declared process/tool scope contains the captured target agent process and its launched tool command, excluding concurrent control activity. AgentSight joins system effects within this scope, and the adapter emits operations carrying those links. AGENTPROF then projects and folds their fields and weights.

Field derivation. Production uses first-match regex rules of the form `KIND:TAG=REGEX`, which developers refine by rerunning AGENTPROF and inspecting unmatched operations. A local LLM tagger runs a quantized 3B-parameter model through `llama.cpp` (Gerganov and contributors 2026) to generate a single-word tag or select from a user-declared task-tag set. Both modes use grammar-constrained decoding and caching. TF-IDF (Salton and Buckley 1988) and K-Means (MacQueen 1967) discover candidate prompt groups without predefined rules, while mapping rules derive fields for structured trajectories, for example `phase:navigate=type:(click|scroll|key)`.

Boundary construction. The Rust inducer counts adjacent action transitions in reference sessions and scores each transition by normalized pointwise mutual information (NPMI), $\text{NPMI}(a, b) = \ln[p(a, b)/(p_L(a)p_R(b))]/-\ln p(a, b)$, with all three probabilities defined over the same transition sample space (Bouma 2009).

Occurrence-weighted one-dimensional k -means with $k = 2$ (MacQueen 1967) initializes at the minimum and maximum scores, assigns distance ties to the lower center, and uses the converged midpoint as a global cutoff. The inducer applies the same procedure to action-changing transitions for a second cutoff. Same-action pairs use the global cutoff, whereas different-action pairs use the smaller cutoff, so the second cutoff can remove but not add a boundary from the global rule. Unseen or below-cutoff transitions start segments whose run-length-compressed action sequences be-

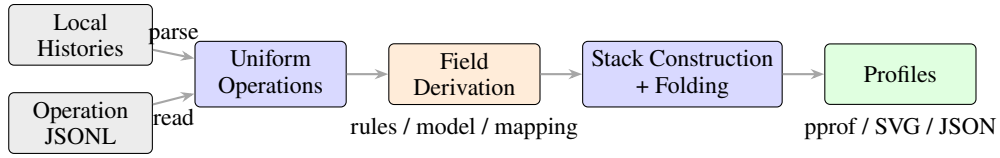


Figure 1: AGENTPROF pipeline. Local histories enter through native history parsing, whereas AgentSight adapters and public datasets enter as newline-delimited JSON operation records. All yield uniform operations before field derivation, stack construction, folding, and profile export.

come stack-frame values. The inducer reads only session identity, input order, and action, not target annotations or resource weights.

An optional reference-calibrated recurrence mode keeps the NPMI score but selects one scalar cutoff that maximizes per-operation B^3 F1 on disjoint reference groups. The mode enumerates score intervals, resolves exact ties toward the smallest cutoff, and never reads target groups. Label-free k -means remains the default.

Profile export. Output formats are pprof protobuf for `go tool pprof -http`, folded stacks for `flamegraph.pl`, standalone SVG flame graphs, and JSON.

Evaluation

Research questions. The evaluation addresses four research questions: **RQ1:** Does semantic profiling improve resource attribution? **RQ2:** Does profiler output correspond to real problems? **RQ3:** How accurate are the tags? **RQ4:** What is the profiling cost? Each RQ subsection pairs its protocol with the strongest answer its evidence supports.

We evaluate three data classes separately. **Real inputs (RQ1)** include 325 local histories (198 Codex, 50 Claude, and 77 Claude subagent) with 183,714 system observations collected over multiple months and a separate 20-task controlled capture suite. **Annotated trajectories (RQ1, RQ3, RQ4)** include 15 real-agent or human web, API, coding, and mobile/GUI execution families mapped into 47,590 operations, plus 405 CodeTraceBench trajectories with 20,866 operations (Lù, Kasner, and Reddy 2024; Deng et al. 2023; Xu et al. 2025; Yao et al. 2022; Li et al. 2023; Qin et al. 2024; Yao et al. 2025; Yang et al. 2024; Li et al. 2024; Lu et al. 2025; Liu et al. 2026b; Lù et al. 2025; Chen et al. 2026b; Wang et al. 2025; Abhyankar, Qi, and Zhang 2025). **Problem-localization benchmarks (RQ2)** provide independent targets over 27,346 operations (Fan et al. 2026; Wang et al. 2026; Chen et al. 2026a). The RQ4 union contains 27,765 operations (Lù et al. 2025; Chen et al. 2026b; Abhyankar, Qi, and Zhang 2025; Wang et al. 2025).

RQ1: Resource Attribution

RQ1 asks whether semantic profiling improves resource attribution beyond what traditional tag-free tools provide.

We first test whether the capture-and-join path assigns system effects within the declared process/tool scope while rejecting concurrent controls. AgentSight 0.2.37 captures 20 real Codex tasks, each paired with one concurrent control,

Method	B^3 P	B^3 R	B^3 F1
Label-free recurrence	0.829	0.534	0.649
Raw-action grouping	0.891	0.388	0.541
Phase only	0.686	0.626	0.654
Action kind	0.948	0.315	0.473
Per session	0.174	1.000	0.296
Per operation	1.000	0.141	0.248

Table 1: RQ1 agreement with human responsibility partitions on 405 CodeTraceBench trajectories. Ordinary B^3 is the standard partition-agreement metric.

and an adapter converts joined effects to operation JSONL for `agentpprof 0.2.37`.

All task and control runs completed. The path recovers 1,520/1,574 in-scope effects with no false positives (100.0% precision, 96.569% recall), rejects all 1,629 control effects, and emits 1,520 samples in 152 stacks while preserving the input and all five predeclared task-category totals.

This establishes scoped capture-and-join accuracy and lossless folding with predeclared rather than inferred task categories.

We next test whether the current operation stack partitions operations into independently annotated responsibility units more faithfully than raw operation identity. We use all 405 failed CodeTraceBench trajectories whose released sources yield the official operation sequence. The population contains 20,866 operations and 2,948 human-verified contiguous stage intervals across OpenHands, mini-SWE-agent, SWE-agent, and Terminus2 (Li et al. 2026). The RQ3 recurrence constructor reads neither target stages nor resource weights. Matched controls start a new group when the raw-action field changes, group by the named phase or action-kind source field, place each session in one group, or make every operation a singleton. Every method receives the same operations and stages.

We use ordinary operation-level B^3 precision, recall, and F1 as the primary metric for partition agreement (Bagga and Baldwin 1998).

Table 1 shows that recurrence raises ordinary B^3 F1 over raw action identity from 0.541 to 0.649. The 0.108 gain has a task-clustered paired-bootstrap 95% interval of [0.087, 0.129] and is positive in all four agent frameworks.

Phase-only reaches 0.654 ordinary B^3 F1, close to recurrence’s 0.649. This experiment therefore supports semantic responsibility partitioning over raw action identity, but not recurrence alone or one universally dominant semantic profile.



Figure 2: Semantic flame graphs of 325 real agent trajectories. Width represents token count (top), summed per-operation elapsed duration (middle), and the number of file operations (bottom). All three aggregate the same operation corpus and differ only in included operations, stack fields, and the additive width measure.

RQ3 separately tests recurrence against phase-only boundaries on OSWorld-Human, where recurrence has a larger advantage. Because this population influenced constructor selection, the result is post-hoc support rather than independent confirmation.

As a separate use case, five declared field hierarchies project the same 13,265 operations from WebLINX, WebShop, APiBank, AgentTrek, SWE-agent, Mind2Web, ToolBench, AndroidControl, and GUI Odyssey. The profiles contain 9 dataset, 57 phase, 226 semantic, 455 action, or 3,757 per-session groups. The semantic hierarchy adds task, operation type, tool, and status to dataset and phase, so across these nine datasets the five depths compare budgets by task, repeated actions across sessions, and individual-session details. Separate adapters map all 15 annotated families listed above into the same operation model without type-specific objects.

Changing only the selection predicate, stack fields, and weight function produces the `tokens`, `time`, and `files` profiles in Figure 2. The `time` and `tokens` views of the same 325 trajectories share only 7/10 top prompt categories (Spearman $\rho = 0.623$), while `network`-tagged prompts rank 8th by time but 93rd by tokens. Recorded span-time and count weights expose distinct bottlenecks; span time is elapsed, not active CPU time, and may include idle or user wait. Together, RQ1 tests effect lineage through scoped controls, conservation through exact totals, and human-partition agreement through ordinary B^3 ; selectable stacks and weights expose different views over the same operations. RQ3 separately evaluates field, group, and boundary construction.

RQ2: Problem Correspondence

RQ2 asks whether target-blind profiles place independently annotated problems in groups a developer can inspect early. We run complete AgentProcessBench, HINTBench, and TraceElephant workloads (Fan et al. 2026; Wang et al. 2026; Chen et al. 2026a). Of HINTBench’s reported 629 trajectories, the released test snapshot used here contains 536. A separate 80-trajectory validation snapshot selects one of 24 field orders before all 536 tests are scored.

Within each workload, AGENTPROF and raw-action grouping use identical operations and precomputed benchmark judge/localizer predictions, differing only in grouping. Predictions, group scores, profiles, fields, and scoring rules are fixed before human/scorer target labels are loaded. AgentProcessBench averages judge votes within each group. For HINTBench and TraceElephant, every root-to-frame stack prefix receives the 95% Wilson lower bound (Wilson 1927) of its member hits, and an operation inherits the maximum score among prefixes containing it, with ties preserved.

Each query is one trajectory containing an annotated target. We report non-interpolated per-query AP, evaluating ties at their shared score threshold, and arithmetic-mean MAP across the 614, 400, and 220 target-bearing queries, respectively (Robertson 2008).

The post-hoc local-only column ranks operations by the benchmark-provided diagnostic score. Local+semantic and local+raw preserve every strict local-score ordering and use the semantic or raw-action group score only to refine exact ties.

In the primary comparison, AGENTPROF improves MAP over raw action on all three complete workloads (Table 2). The gains are +.016 [.005,.027], +.171 [.154,.188], and +.109 [.077,.142], using 10,000 paired resamples of trajectory clus-

Workload	Sem.	Raw	Local+S	Local+R	Local
AgentProcessBench	.789	.773	.896	.893	.863
HINTBench	.452	.281	.545	.506	.411
TraceElephant	.230	.121	.322	.249	.209

Table 2: Standard per-query AP averaged as MAP after scoring every operation. Semantic versus raw action is the primary comparison. Local+semantic, local+raw, and local-only are a post-hoc mechanism analysis on the same complete populations. Higher is better.

ters within benchmark-defined strata.

In this post-hoc mechanism analysis, local+semantic MAP is .896/.545/.322, above local-only and semantic-only on all three workloads. Relative to the identically composed local-plus-raw tie-breaker, it gains +.039 [.029,.049] on HINT-Bench and +.073 [.050,.097] on TraceElephant. AgentProcessBench does not distinguish the two refinements (+.003 [-.0005,.0069]).

Semantic grouping improves MAP over matched raw-action grouping on all three complete workloads. The adaptive local-first results support semantic recurrence as a tie-breaking refinement, not a replacement, of operation-local diagnostic scores, but are not untouched confirmation.

RQ3: Tag Accuracy

RQ3 evaluates predeclared field derivation against independent annotations under each protocol’s stated input boundary. Literal task and action fields use accuracy and macro-F1 (the unweighted mean of per-class F1) (Lewis et al. 2004). Partition-valued task, phase, and group fields use V-measure (Rosenberg and Hirschberg 2007) or ordinary B³, and adjacent group boundaries use exact precision, recall, and F1 (Ruokolainen et al. 2016). Literal labels, permutation-invariant partitions, and adjacent boundaries are distinct outputs. A *tag* is an operation field consumed by AGENTPROF’s pluggable field interface. Whether run through the CLI or a standalone adapter, a backend maps its output to this field, from which AGENTPROF derives structural groups and boundaries. AGENTPROF folds predicted OSWorld groups with their original additive weights.

Table 1 evaluates the deterministic phase field structurally: phase-only grouping reaches 0.654 ordinary B³ F1 against 2,948 human stages over 20,866 operations. Below we test literal labels, other partitions, and boundaries.

We test all 287 OSWorld-Human task-instance sessions (Abhyankar, Qi, and Zhang 2025) with complete group annotations: 3,978 operations, 3,691 adjacent pairs, and 2,042 groups. For the supervised Bernoulli Naive Bayes predictor (McCallum and Nigam 1998), we fix the model family, nine input fields, adjacent-pair features, and training procedure. Each session-held-out cross-validation fold fits its parameters and threshold on training sessions and predicts every held-out session once. Predicted boundaries become an ordinary group field that AGENTPROF projects and folds. With the same five folds, label-free recurrence uses only the other sessions’ visible action transitions. Because

Method	Prec.	Recall	Bound. F1	B ³ F1
Supervised predictor	0.700	0.782	0.739	0.816
Reference-calibrated recurrence	0.610	0.922	0.734	0.801
Label-free recurrence	0.592	0.799	0.680	0.786
Always boundary	0.476	1.000	0.645	0.678
Action change	0.386	0.626	0.477	0.659
Phase change	0.441	0.268	0.334	0.665

Table 3: RQ3 boundary and partition agreement on OSWorld-Human. Boundary F1 scores exact adjacent-pair decisions. B³ scores predicted–human partitions.

we designed the recurrence rule after inspecting earlier results on this corpus, these results are development evidence rather than independent cross-family confirmation. Optional reference-calibrated recurrence uses training operations and group annotations to fit one NPMI cutoff, while held-out folds contribute only visible actions until predictions are fixed.

Controls place a boundary after every operation, visible action change, or visible phase change. We also report ordinary per-operation B³ partition F1 (Bagga and Baldwin 1998), whose predicted–human group overlap captures merging and fragmentation.

The supervised predictor reaches 0.739 boundary F1 and 0.816 B³ F1 versus 0.645 and 0.678 for the strongest controls. Label-free recurrence reaches 0.680 and 0.786, while reference-calibrated recurrence reaches 0.734 and 0.801 but requires group annotations that the default label-free method does not use. Every method conserves the total projected weight.

For target-blind TF-IDF/K-Means, V-measure (Rosenberg and Hirschberg 2007) over operation assignments reaches 0.557 on nine Mind2Web sessions (49 operations) and 0.815 on 100 ScienceWorld sessions (2,504 operations) (Deng et al. 2023; Wang et al. 2022). Every operation receives a prediction in both datasets, whereas a constant tag yields V-measure 0. Here, V-measure tests partitions, not literal names.

For task-family tags, a fixed evaluation-only Qwen3.6-27B llama.cpp backend (distinct from the 3B CLI backend) classifies all 1,012 AgentBoard goals into nine families using only goals and declared family descriptions (Qwen Team 2026; Ma et al. 2024). It reaches 0.695 macro-F1 and 0.733 accuracy versus a majority-class baseline at 0.044 and 0.248, with identical assignments across three runs.

For action tags, a standalone backend-level adapter applies the fixed Qwen3.6-27B closed-label configuration to only the current thought/action and eight definitions for all 2,737 published annotations from 120 AutoCodeRover, OpenHands, and RepairAgent trajectories (Qwen Team 2026; Bouzenia and Pradel 2025; Sajadi et al. 2026). The model reaches 0.498 macro-F1 and 0.628 accuracy versus a majority-class baseline at 0.061 and 0.323. Its 0.437 macro-F1 gain has a whole-trajectory 95% bootstrap interval of [0.380, 0.494], and both complete runs agree exactly. Thirty-nine AutoCodeRover inputs literally contain the target word `locate`. Excluding them leaves 0.490 macro-F1 and 0.622 accuracy, so the positive comparison to majority is unchanged.

Workload	Ops	Semantic (s)	Raw action (s)	Peak RSS (MiB)
AgentRewardBench	729	0.04	0.03	19.0
SATraj-OS	4,285	0.17	0.14	73.5
OSWorld-Human	6,010	0.25	0.21	108.4
AgentNet	16,741	0.71	0.58	278.4
Union	27,765	1.17	0.99	464.5

Table 4: RQ4 profile-construction cost. Times are three-run medians, and resident set size (RSS) is the largest peak observed for the semantic profile.

Thus, on the named public populations, RQ3 establishes target-blind task partitions at V-measure 0.557 and 0.815, phase structure at 0.654 ordinary B³ F1, task-family assignment at 0.695 macro-F1, action assignment at 0.498 macro-F1, and label-free group boundaries at 0.680 boundary F1 and 0.786 B³ F1.

RQ4: Profiling Cost

RQ4 measures the time to parse fixed-field operation JSONL, construct stacks, fold weights, and serialize the profile. The measurement excludes capture, source adaptation, and field/tag generation. Across four complete public workloads and their union, we compare the fixed semantic hierarchy `project, agent, task, phase, op, tool, status` with the raw hierarchy `project, agent, action, status` on identical inputs over three runs. Measurements use `agentpprof 0.2.37` on a 24-core Intel Core Ultra 9 285K with 125 GiB RAM and Linux 6.15.11.

Both time curves are monotonic across the five input sizes. The semantic curve has a descriptive slope of 0.0422 ms per operation ($R^2 = 0.9997$), reaching 23,731 operations/s on the union. RQ4 therefore finds that, for 27,765 operations, semantic construction completes in 1.17 seconds using 464.5 MiB, takes 180 ms (18.2%) more time, and uses 6.0 MiB (1.3%) more peak RSS than raw-action grouping.

Evidence synthesis. The four RQs validate different links in one profiling chain. RQ1 tests scoped effect lineage and conservation before measuring partition agreement, while RQ2 tests independently annotated problem ranking and semantic refinement of operation-local evidence. RQ3 tests the pluggable field interface through literal task/action labels and structural partitions and boundaries, and RQ4 tests fixed-input profile construction. Together, they connect cross-layer operations and selectable operation stacks to the attribution, localization, semantic-field, and cost questions that motivate agent profiling.

Scope and Limitations

RQ1 covers 20 fixed tasks under declared process/tool scope and all 405 failed CodeTraceBench trajectories (Li et al. 2026) reconstructable as official operation sequences. Because this corpus influenced constructor selection, its result is post-hoc support, not independent confirmation.

RQ2 evaluates all target-bearing queries in the three named released populations. Its local-first mechanism analysis is adaptive to those same populations and does not replace the primary semantic-versus-raw comparison.

RQ3 tests CodeTraceBench phase structure, Mind2Web/ScienceWorld partitions, a named backend over nine task families and eight action tags, and session-held-out OSWorld-Human boundaries. These measurements cover the named populations and declared tag sets.

RQ4 covers parsing fixed-field operation input, stack construction, folding, and serialization. Capture, source adaptation, field/tag generation, and live agent overhead remain outside that construction-cost measurement.

Related Work

Agent observability. Platforms aggregate spans, metadata, and evaluations (LangChain 2024; Langfuse 2024; OpenTelemetry 2024; Datadog 2025). Datadog Patterns and LangSmith Insights derive cross-trace hierarchies and roll up metrics, NeMo profiles instrumented workflows, and OpenTelemetry Profiles links profiles to traces (Datadog 2026; LangChain 2026; NVIDIA 2026; OpenTelemetry 2026). These establish grouping, rollups, profile naming and compatibility, and trace linkage. AGENTPROF’s residual capability is their conjunction in one operation corpus: joining agent actions to process, file, and network effects while conserving arbitrary additive measures and supporting selectable query-time field stacks.

Cross-run agent analysis. TraceProbe supplies resource-aware process profiles, while Graphectory builds per-trajectory process graphs with aggregated phase patterns (Shu et al. 2026; Liu et al. 2026a). Hodoscope compares behavior distributions to direct human review, while TraceGraph builds shared decision landscapes that guide recovery (Zhong, Saxena, and Raghunathan 2026; Nian et al. 2026). Together, these systems establish canonical actions, recurring cross-run structure, and analysis-to-decision precedents. AGENTPROF complements them with source-linked heterogeneous effects, conservation of user-selected additive measures, and multiple operation-stack projections over the same histories.

Profiling and diagnosis. `pprof` and flame graphs fold runtime stacks, whereas Pivot Tracing dynamically selects and groups measurements across causally related events (Google 2024; Gregg 2011; Mace, Roelke, and Fonseca 2015). `AgentRx` diagnoses failures, `CodeTracer` reconstructs per-run states, and localization benchmarks score faulty steps, agents, or expert-supported perspectives (Barke et al. 2026; Li et al. 2026; Fan et al. 2026; Wang et al. 2026; Chen et al. 2026a; In et al. 2026). AGENTPROF applies the profiler’s population-level principle to agents by treating completed trajectories as samples and using query-time operation stacks to fold recurring semantic responsibility without code identity or runtime nesting.

Conclusion

Agent observability needs profiling, not only debugging. AGENTPROF represents heterogeneous agent actions and sys-

tem effects as weighted operations and folds selectable query-time operation stacks across runs. On complete real and public workloads, its profiles preserve linked effects, improve human-stage partition agreement and problem-localization MAP over raw action, and recover OSWorld-Human groups at 0.680 boundary F1 and 0.786 B³ F1. A 27,765-operation profile builds in 1.17 seconds, making population-level profiling a practical complement to per-run debugging.

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